MinerU: An Open-Source Solution for Precise Document Content Extraction

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Abstract

Document content analysis has been a crucial research area in computer vision. Despite significant advancements in methods such as OCR, layout detection, and formula recognition, existing open-source solutions struggle to consistently deliver high-quality content extraction due to the diversity in document types and content. To address these challenges, we present MinerU, an open-source solution for high-precision document content extraction. MinerU leverages the sophisticated PDF-Extract-Kit models to extract content from diverse documents effectively and employs finely-tuned preprocessing and postprocessing rules to ensure the accuracy of the final results. Experimental results demonstrate that MinerU consistently achieves high performance across various document types, significantly enhancing the quality and consistency of content extraction. The MinerU open-source project is available at `https://github.com/opendatalab/MinerU`

1 Introduction

The release of ChatGPT [23, 24] at the end of 2022 ignites a wave of interest in the research and application of large language models (LLMs) [15, 27, 29, 41, 6, 30, 7, 5, 1, 12]. Central to training high-quality LLMs is the acquisition and construction of high-quality data. As LLMs rapidly evolve, data from internet web pages is becoming insufficient to support further improvements in model training. Document data, which contains a wealth of knowledge, emerges as a crucial resource for enhancing LLMs. The introduction and development of Retrieval-Augmented Generation (RAG) [14, 26, 2, 8] in 2023 further intensify the demand for high-quality document extraction in both industry and academia.

Currently, there are four main technical approaches to document content extraction:

OCR-based Text Extraction. This approach uses OCR models to directly extract text from documents. While feasible for plain text documents, it introduces significant noise when documents contain images, tables, formulas, and other elements, rendering it unsuitable for high-quality data extraction.

Library-based Text Parsing. For non-scanned documents, open-source Python libraries such as PyMuPDF can directly read content without invoking OCR, offering faster and more accurate text results. However, this approach fails when documents contain formulas, tables, and other elements.

Multi-Module Document Parsing. This approach employs various document parsing models to process document images in multiple stages. Initially, layout detection algorithms identify different

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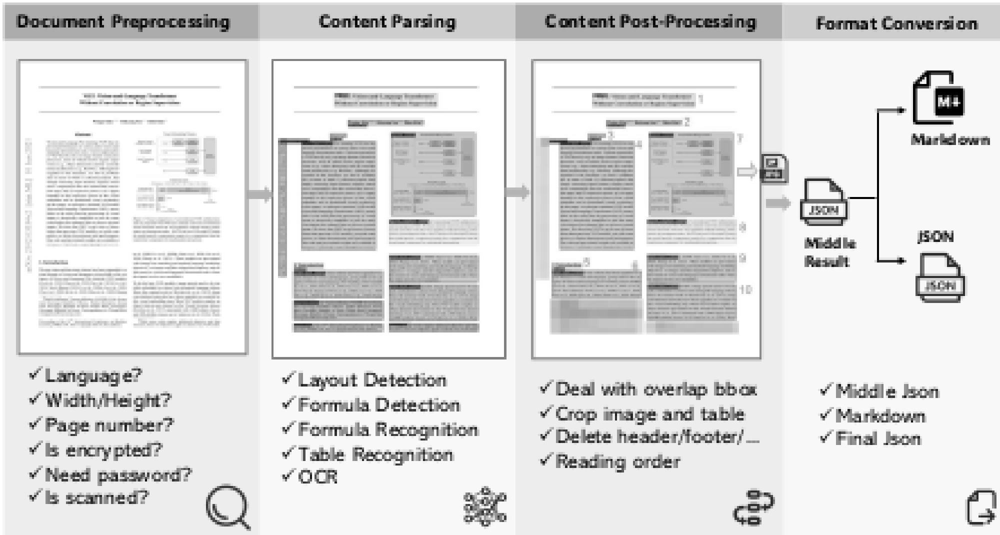

Figure 1: Overview of the MinerU framework processing workflow.

types of regions, such as images, image captions, tables, table captions, headings, and text. Subsequently, different recognizers are applied to these specific regions. For instance, OCR is used for text and headings, formula recognition models handle formulas, and table recognition models process tables. Although this method is theoretically capable of producing high-quality document results, existing open-source models often focus solely on academic papers and perform poorly on diverse document types, including textbooks, exam papers, research reports, and newspapers.

End-to-End MLLM Document Parsing. With the advancement of multimodal large language models (MLLMs), numerous models for document content extraction emerge, such as Donut [13], Nougat [4], Kosmos-2.5 [21], Vary [34], Vary-toy [35], mPLUG-DocOwl-1.5 [9], mPLUG-DocOwl2 [10], Fox [17], and GOT [36]. These models benefit from continuously optimized encoders (e.g., SwinTransformer [20], ViTDet [16]) and decoders (e.g., mBART [18], Qwen2-0.5B [38]) as well as data engineering, gradually improving extraction performance. However, they still face challenges related to data diversity and high inference costs.

To better extract diverse documents while ensuring low inference costs and high inference quality, we propose MinerU, an all-in-one document extraction tool. MinerU's primary technical approach is based on the multi-module document parsing strategy. Unlike existing document parsing algorithms, MinerU leverages various open-source models from the PDF-Extract-Kit[3], which are trained on diverse real-world documents to achieve high-quality results in tasks involving complex layouts and intricate formulas. After obtaining the positions and recognition content of different regions from the models, MinerU employs a tailored processing workflow to ensure the accuracy of the results.

Using MinerU for document extraction offers several advantages:

- **Adaptability to Diverse Document Layouts**: Supports a wide range of document types, including but not limited to academic papers, textbooks, exam papers, and research reports.
- **Content Filtering**: Allows filtering of irrelevant regions such as headers, footers, footnotes, and side notes, enhancing document readability.
- **Accurate Segmentation**: Combines model-based and rule-based post-processing for paragraph recognition, enabling cross-column and cross-page paragraph merging.
- **Robust Page Element Recognition**: Accurately distinguishes between formulas, tables, images, text blocks, and their respective captions.

[3] https://github.com/opendatalab/PDF-Extract-Kit

2 MinerU Framework

As shown in Figure 1, MinerU processes diverse user-input PDF documents and converts them into desired machine-readable formats (Markdown or JSON) through a series of steps. Specifically, the processing workflow of MinerU is divided into four stages:

Document Preprocessing. This stage uses PyMuPDF[4] to read PDF files, filter out unprocessable files (e.g., encrypted files), and extract PDF metadata, including the document's parseability (categorized into parseable and scanned PDFs), language type, and page dimensions.

Document Content Parsing. This stage employs the PDF-Extract-Kit, a high-quality PDF document extraction algorithm library, to parse key document contents. It begins with layout analysis, including layout and formula detection. Different recognizers are then applied to various regions: OCR [28, 19] for text and titles, formula recognition [3, 25, 33] for formulas, and table recognition [37] for tables.

Document Content Post-Processing. Based on the outputs from the second stage, this stage removes invalid regions, stitches content according to regional positioning information, and ultimately obtains the positioning, content, and sorting information for different document regions.

Format Conversion. Based on the results of document post-processing, various formats required by users, such as Markdown, can be generated for subsequent use.

2.1 Document Preprocessing

PDF document preprocessing has two main objectives: first, to filter out unprocessable PDFs, such as non-PDF files, encrypted documents, and password-protected documents. Second, to obtain PDF metadata for subsequent use. The acquisition of PDF metadata includes the following aspects:

- **Language Identification**: Currently, MinerU identifies and processes only Chinese and English documents. The language type needs to be specified as a parameter when performing OCR, and the quality of processing for other languages is not guaranteed.
- **Content Garbled Detection**: Some text-based PDFs contain text that appears garbled when copied. Such PDFs need to be identified in advance so that OCR can be used for text recognition in the next step.
- **Scanned PDF Identification**: For text-based PDFs (as opposed to scanned PDFs), MinerU directly uses PyMuPDF for text extraction. However, for scanned PDFs, OCR needs to be enabled. Scanned PDFs are identified based on characteristics such as a larger image area compared to the text area, sometimes covering the entire PDF page, and an average text length per page close to zero.
- **Page Metadata Extraction**: Extracts document metadata such as total page count, page dimensions (width and height), and other relevant attributes.

2.2 Document Content Parsing

In the document parsing stage, MinerU uses the PDF-Extract-Kit model library to detect different types of regions and recognize the corresponding region contents (OCR, formula recognition, table recognition, etc.). PDF-Extract-Kit is an algorithm library for PDF parsing, containing various state-of-the-art (SOTA) open-source PDF document parsing algorithms. Unlike other open-source algorithm libraries, PDF-Extract-Kit aims to build a model library that ensures accuracy and speed when dealing with diverse data in real-world scenarios. When the SOTA open-source algorithms in a specific field fail to meet practical needs, PDF-Extract-Kit employs data engineering to construct high-quality, diverse datasets for further model fine-tuning, thereby significantly enhancing the model's robustness to varied data. The current version of MinerU[5] utilizes five models: layout detection, formula detection, table recognition, formula recognition and OCR.

[4] https://github.com/pymupdf/PyMuPDF
[5] Current version: v0.8.1

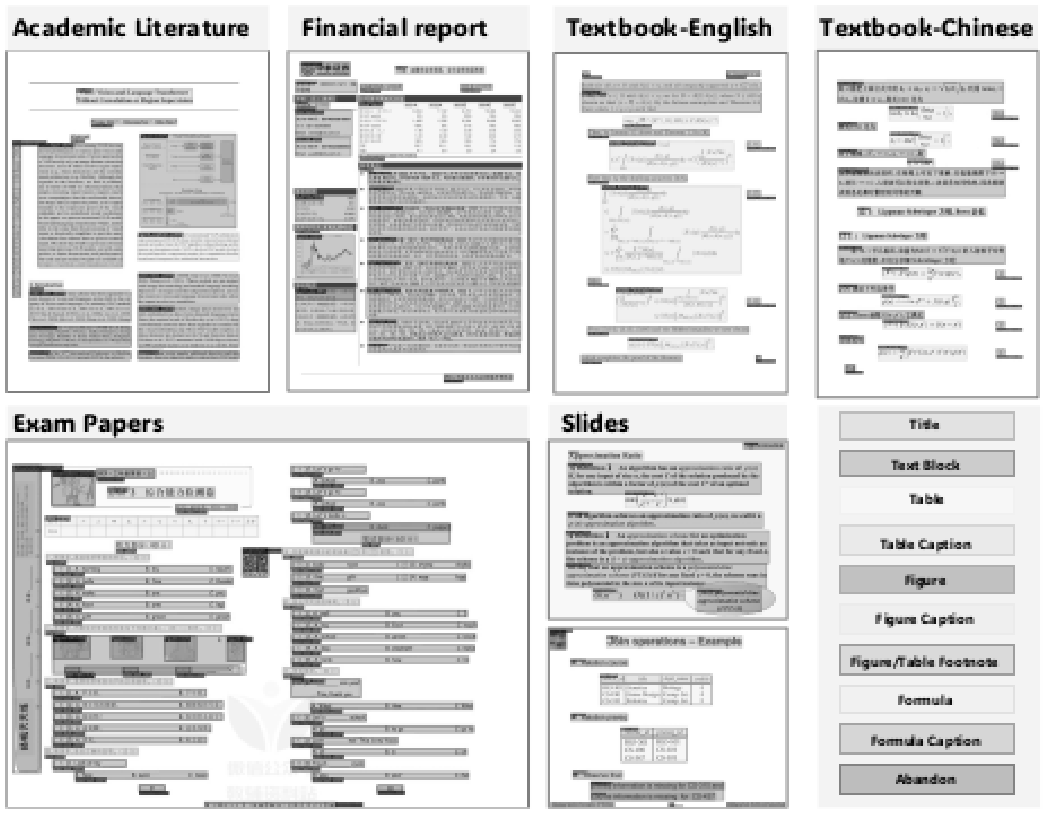

Figure 2: High-quality layout detection results on diverse documents.

2.2.1 Layout Analysis

Layout analysis is the crucial first step in document parsing, aiming to distinguish different types of elements and their corresponding regions on a page. Existing layout detection algorithms [11, 39] perform well on paper-type documents but struggle with diverse documents such as textbooks and exam papers. Therefore, PDF-Extract-Kit constructs a diverse layout detection training set and trains high-quality models for document region extraction.

The data engineering-based model training approach is as follows:

- **Diverse Data Selection**: Collects diverse PDF documents, clusters them based on visual features, and samples data from different cluster centers to obtain an initial diverse document dataset. The categories include scientific papers, general books, textbooks, exam papers, magazines, PPTs, research reports, etc.
- **Data Annotation**: Categorizes the layout annotation types involved in the document components, including titles, body paragraphs, images, image captions, tables, table captions, image table notes, inline formulas, formula labels, and discard types (such as headers, footers, page numbers, and page notes). Detailed annotation standards are established for each type, and approximately 21K data points are annotated as the training set.
- **Model Training**: Fine-tunes the model for the Layout Detection task based on the layout detection models [11, 31]. The number of classes parameter is modified to align with our categorized layout types.
- **Iterative Data Selection and Model Training**: During model iteration, partitions a portion of the data as a validation set and uses its results to guide the focus of subsequent data iterations. If a specific category from a particular source of PDF documents scores low, the sampling weight for PDF pages containing that specific category from that source is increased in the next iteration, thereby more efficiently iterating the data and model.